Business Problem:

Business Problem: Predicting Driver Churn for Ola

Company Overview: Ola is a leading ride-sharing platform focused on delivering reliable, affordable, and convenient urban transportation across India and beyond.

Core Challenge: One of Ola's persistent operational challenges is the high churn rate among drivers. Retaining drivers is critical to maintaining service quality, reducing recruitment costs, and ensuring platform stability.

Objective: To predict driver churn using historical driver data, enabling Ola to take proactive retention measures and improve driver satisfaction.

Data-Driven Approach

- Analyze driver behavior, performance metrics, and engagement patterns.
- Identify key indicators that correlate with churn (e.g., ride frequency, earnings, ratings).
- Use ensemble learning techniques (e.g., Random Forest, XGBoost) to build predictive models that outperform single algorithms.

Business Impact:

- Early identification of at-risk drivers allows targeted interventions (e.g., incentives, support).
- Enhances driver retention, reduces operational costs, and improves overall service reliability.
- Strengthens Ola's competitive edge by maintaining a stable and satisfied driver base.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
```

1. Basic Metrics:

```
df_ola = pd.read_csv('C:\\Users\\njain121\\OneDrive - UHG\\Neha\\
Scalar\\CaseStudy\\Ola\\ola_driver_scaler.csv')
df_ola.head()
```

```
   Unnamed: 0    MMM-YY  Driver_ID   Age  Gender City  Education_Level
\
0           0  01/01/19          1  28.0     0.0  C23                2

1           1  02/01/19          1  28.0     0.0  C23                2

2           2  03/01/19          1  28.0     0.0  C23                2

3           3  11/01/20          2  31.0     0.0   C7                2
```

```
4           4  12/01/20          2  31.0     0.0   C7                 2


   Income Dateofjoining LastWorkingDate  Joining Designation  Grade  \
0   57387      24/12/18             NaN                    1      1
1   57387      24/12/18             NaN                    1      1
2   57387      24/12/18        03/11/19                    1      1
3   67016      11/06/20             NaN                    2      2
4   67016      11/06/20             NaN                    2      2

   Total Business Value  Quarterly Rating
0               2381060                 2
1               -665480                 2
2                     0                 2
3                     0                 1
4                     0                 1
```

```
df_ola.shape
```

```
(19104, 14)
```

```
df_ola.info()  ## details of the dataframe
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 19104 entries, 0 to 19103
Data columns (total 14 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   Unnamed: 0            19104 non-null  int64
 1   MMM-YY                19104 non-null  object
 2   Driver_ID             19104 non-null  int64
 3   Age                   19043 non-null  float64
 4   Gender                19052 non-null  float64
 5   City                  19104 non-null  object
 6   Education_Level       19104 non-null  int64
 7   Income                19104 non-null  int64
 8   Dateofjoining         19104 non-null  object
 9   LastWorkingDate       1616 non-null   object
 10  Joining Designation   19104 non-null  int64
 11  Grade                 19104 non-null  int64
 12  Total Business Value  19104 non-null  int64
 13  Quarterly Rating      19104 non-null  int64
dtypes: float64(2), int64(8), object(4)
memory usage: 2.0+ MB
```

Observations on Data

```
df_ola.describe(include='all')
```

```
          Unnamed: 0    MMM-YY     Driver_ID           Age
Gender  \
count   19104.000000     19104  19104.000000  19043.000000
19052.000000
unique           NaN        24           NaN           NaN
NaN
top              NaN  01/01/19           NaN           NaN
NaN
freq             NaN      1022           NaN           NaN
NaN
mean     9551.500000       NaN   1415.591133     34.668435
0.418749
std      5514.994107       NaN    810.705321      6.257912
0.493367
min         0.000000       NaN      1.000000     21.000000
0.000000
25%      4775.750000       NaN    710.000000     30.000000
0.000000
50%      9551.500000       NaN   1417.000000     34.000000
0.000000
75%     14327.250000       NaN   2137.000000     39.000000
1.000000
max     19103.000000       NaN   2788.000000     58.000000
1.000000

         City  Education_Level         Income Dateofjoining
LastWorkingDate  \
count   19104     19104.000000   19104.000000         19104
1616
unique     29              NaN            NaN           869
493
top       C20              NaN            NaN      23/07/15
29/07/20
freq     1008              NaN            NaN           192
70
mean      NaN         1.021671   65652.025126           NaN
NaN
std       NaN         0.800167   30914.515344           NaN
NaN
min       NaN         0.000000   10747.000000           NaN
NaN
25%       NaN         0.000000   42383.000000           NaN
NaN
50%       NaN         1.000000   60087.000000           NaN
NaN
75%       NaN         2.000000   83969.000000           NaN
NaN
max       NaN         2.000000  188418.000000           NaN
NaN
```

```
        Joining Designation         Grade  Total Business Value  \
count          19104.000000  19104.000000          1.910400e+04
unique                  NaN           NaN                   NaN
top                     NaN           NaN                   NaN
freq                    NaN           NaN                   NaN
mean               1.690536      2.252670          5.716621e+05
std                0.836984      1.026512          1.128312e+06
min                1.000000      1.000000         -6.000000e+06
25%                1.000000      1.000000          0.000000e+00
50%                1.000000      2.000000          2.500000e+05
75%                2.000000      3.000000          6.997000e+05
max                5.000000      5.000000          3.374772e+07

        Quarterly Rating
count       19104.000000
unique               NaN
top                  NaN
freq                 NaN
mean            2.008899
std             1.009832
min             1.000000
25%             1.000000
50%             2.000000
75%             3.000000
max             4.000000
```

```
df_ola.isna() ##Check for the missing values
```

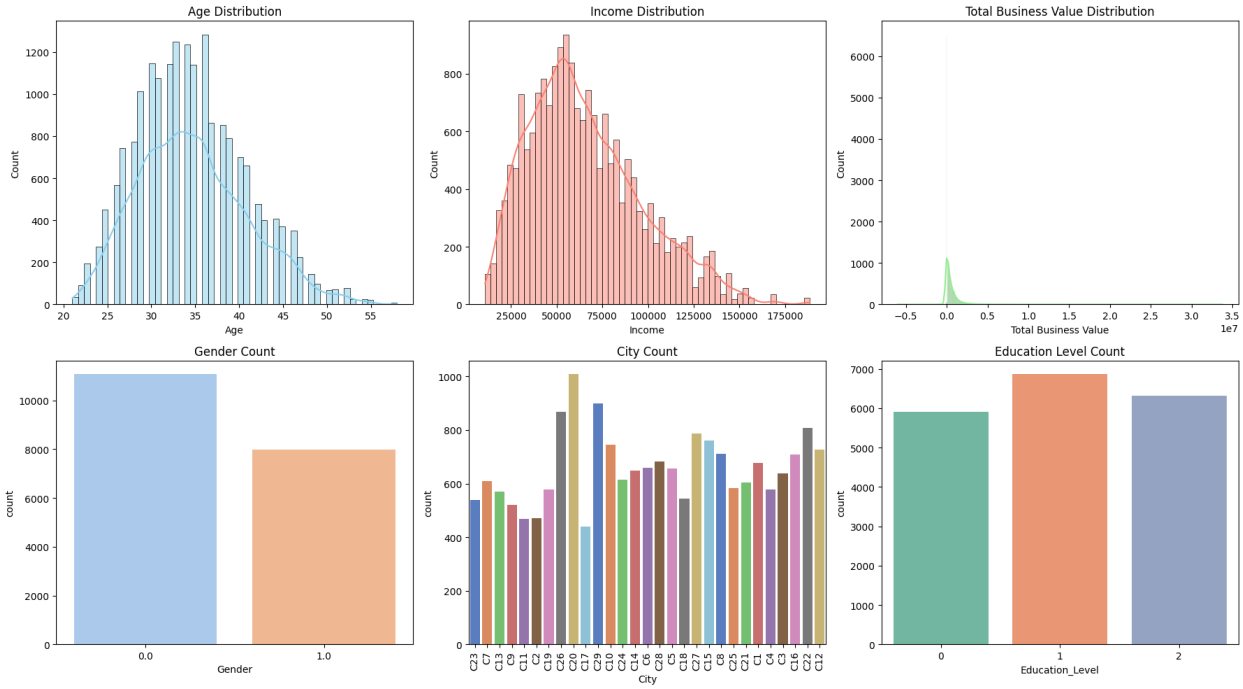
```
       Unnamed: 0  MMM-YY  Driver_ID    Age  Gender   City
Education_Level  \
0           False   False      False  False   False  False
False
1           False   False      False  False   False  False
False
2           False   False      False  False   False  False
False
3           False   False      False  False   False  False
False
4           False   False      False  False   False  False
False
...           ...     ...        ...    ...     ...    ...
...
19099       False   False      False  False   False  False
False
19100       False   False      False  False   False  False
False
19101       False   False      False  False   False  False
False
19102       False   False      False  False   False  False
```

```
False
19103       False   False      False  False   False  False
False

       Income  Dateofjoining  LastWorkingDate  Joining Designation
Grade  \
0       False          False             True                False
False
1       False          False             True                False
False
2       False          False            False                False
False
3       False          False             True                False
False
4       False          False             True                False
False
...       ...            ...              ...                  ...
...
19099   False          False             True                False
False
19100   False          False             True                False
False
19101   False          False             True                False
False
19102   False          False             True                False
False
19103   False          False             True                False
False

       Total Business Value  Quarterly Rating
0                     False             False
1                     False             False
2                     False             False
3                     False             False
4                     False             False
...                     ...               ...
19099                 False             False
19100                 False             False
19101                 False             False
19102                 False             False
19103                 False             False

[19104 rows x 14 columns]
```

Univariate Analysis

```
warnings.filterwarnings('ignore')
# Set up the plot grid
```

```
fig, axes = plt.subplots(2, 3, figsize=(18, 10))

# Continuous variables: Age, Income, Total Business Value
sns.histplot(df_ola['Age'].dropna(), kde=True, ax=axes[0, 0],
color='skyblue')
axes[0, 0].set_title('Age Distribution')

sns.histplot(df_ola['Income'].dropna(), kde=True, ax=axes[0, 1],
color='salmon')
axes[0, 1].set_title('Income Distribution')

sns.histplot(df_ola['Total Business Value'].dropna(), kde=True,
ax=axes[0, 2], color='lightgreen')
axes[0, 2].set_title('Total Business Value Distribution')

# Categorical variables: Gender, City, Education_Level
sns.countplot(x='Gender', data=df_ola, ax=axes[1, 0],
palette='pastel')
axes[1, 0].set_title('Gender Count')

sns.countplot(x='City', data=df_ola, ax=axes[1, 1], palette='muted')
axes[1, 1].set_title('City Count')
axes[1, 1].tick_params(axis='x', rotation=90)

sns.countplot(x='Education_Level', data=df_ola, ax=axes[1, 2],
palette='Set2')
axes[1, 2].set_title('Education Level Count')

plt.tight_layout()
plt.show()
plt.close()
```



Insights

Continuous Variables

1. Age Distribution
- Most drivers fall in the 25–40 years range, indicating Ola's driver base is relatively young.
- Very few drivers are above 50, suggesting limited participation from older age groups.
1. Income Distribution
- Income is right-skewed, with most drivers earning between ₹40,000–₹90,000.
- A small number of drivers earn significantly higher, which could represent high-performing or long-tenured drivers.
1. Total Business Value
- Extremely skewed with heavy outliers, meaning a few drivers contribute disproportionately to Ola's revenue.
- Many drivers have zero or very low business value, possibly indicating inactivity or churn risk.

Categorical Variables

1. Gender Count
- One category dominates (likely male), showing gender imbalance in the driver workforce.
- This could be an area for diversity initiatives.
1. City Count
- Driver distribution varies widely across cities, with some cities having a much larger driver base.
- Indicates regional concentration of Ola's operations.

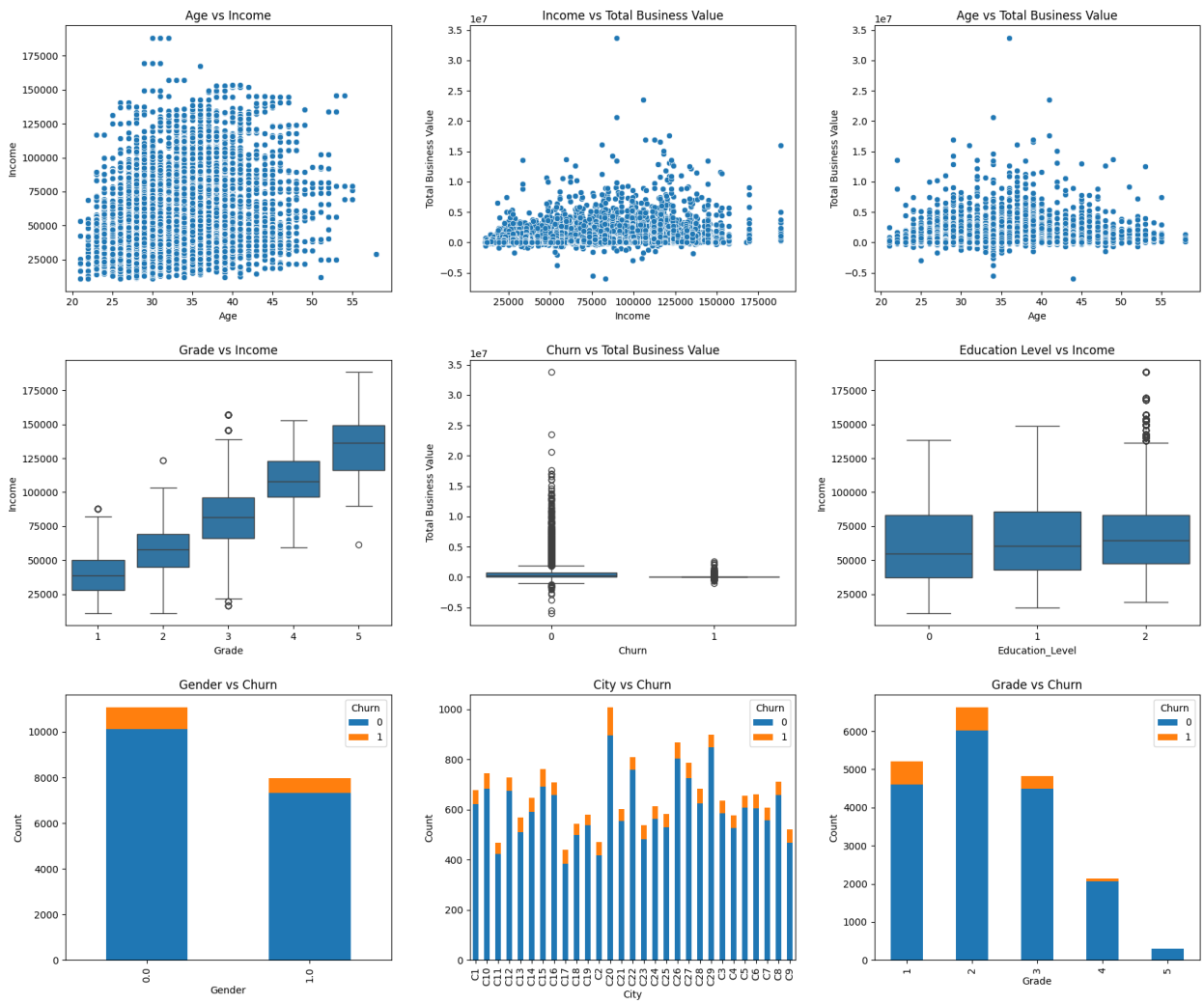
1. Education Level Count

- Most drivers have education level 1 or 2, suggesting a workforce with basic to intermediate education.
- Higher education levels are rare, which may influence training and engagement strategies.

Bivariate Analysis

```
# Create churn column based on LastWorkingDate
df_ola['Churn'] = df_ola['LastWorkingDate'].notna().astype(int)

# Set up the plot grid
fig, axes = plt.subplots(3, 3, figsize=(18, 15))
fig.tight_layout(pad=5.0)

# Scatter plots (continuous-continuous)
sns.scatterplot(data=df_ola, x='Age', y='Income', ax=axes[0, 0])
axes[0, 0].set_title('Age vs Income')

sns.scatterplot(data=df_ola, x='Income', y='Total Business Value',
ax=axes[0, 1])
axes[0, 1].set_title('Income vs Total Business Value')

sns.scatterplot(data=df_ola, x='Age', y='Total Business Value',
ax=axes[0, 2])
axes[0, 2].set_title('Age vs Total Business Value')

# Boxplots (categorical-continuous)
sns.boxplot(data=df_ola, x='Grade', y='Income', ax=axes[1, 0])
axes[1, 0].set_title('Grade vs Income')

sns.boxplot(data=df_ola, x='Churn', y='Total Business Value',
ax=axes[1, 1])
axes[1, 1].set_title('Churn vs Total Business Value')

sns.boxplot(data=df_ola, x='Education_Level', y='Income', ax=axes[1,
2])
axes[1, 2].set_title('Education Level vs Income')

# Stacked bar plots (categorical-categorical)
crosstab_gender_churn = pd.crosstab(df_ola['Gender'], df_ola['Churn'])
crosstab_gender_churn.plot(kind='bar', stacked=True, ax=axes[2, 0])
axes[2, 0].set_title('Gender vs Churn')
axes[2, 0].set_xlabel('Gender')
axes[2, 0].set_ylabel('Count')

crosstab_city_churn = pd.crosstab(df_ola['City'], df_ola['Churn'])
crosstab_city_churn.plot(kind='bar', stacked=True, ax=axes[2, 1])
axes[2, 1].set_title('City vs Churn')
```

```
axes[2, 1].set_xlabel('City')
axes[2, 1].set_ylabel('Count')

crosstab_grade_churn = pd.crosstab(df_ola['Grade'], df_ola['Churn'])
crosstab_grade_churn.plot(kind='bar', stacked=True, ax=axes[2, 2])
axes[2, 2].set_title('Grade vs Churn')
axes[2, 2].set_xlabel('Grade')
axes[2, 2].set_ylabel('Count')

# Save the figure
plt.show()
plt.close()
```



Insights

Scatter Plots (Continuous vs Continuous)

1. Age vs Income

- Income does not strongly correlate with age; drivers across all ages have similar income ranges.
- Slight clustering around ages 25–40 with mid-level incomes.

1. Income vs Total Business Value

- No clear linear relationship; some drivers with moderate income generate very high business value.
- Indicates that income alone is not a strong predictor of business contribution.

1. Age vs Total Business Value

- Similar pattern: business value varies widely across age groups.
- Outliers dominate, suggesting a few drivers contribute disproportionately.

Boxplots (Categorical vs Continuous)

1. Grade vs Income

- Higher grades generally correspond to higher median income.
- Significant overlap between grades, but Grade 4–5 drivers show higher income ranges.

1. Churn vs Total Business Value

- Churned drivers tend to have lower business value compared to active drivers.
- Indicates that high-value drivers are less likely to churn.

1. Education Level vs Income

- Higher education levels slightly correlate with higher income.
- However, the difference is not very pronounced.

Stacked Bar Charts (Categorical vs Categorical)

1. Gender vs Churn

- Majority of churn cases occur in the dominant gender group (likely male), but proportionally churn seems similar across genders.

1. City vs Churn

- Churn distribution varies by city; some cities have higher churn rates relative to their driver base.
- Could indicate regional operational challenges.

1. Grade vs Churn

- Lower grades show higher churn counts.
- Higher-grade drivers are more stable, reinforcing the link between performance and retention.

Perform data preprocessing

Duplicate and Missing value check

```
df_ola.drop_duplicates(inplace=True)
df_ola.shape
```

```
(19104, 15)
```

```python
# Check for duplicates based on a subset of columns
subset_cols = ['Driver_ID', 'Dateofjoining', 'City']
duplicates = df_ola[df_ola.duplicated(subset=subset_cols, keep=False)]

# Count and show sample duplicates
duplicate_count = duplicates.shape[0]
sample_duplicates = duplicates.head(10)

print(f"Number of duplicate entries based on {subset_cols}:
{duplicate_count}\n")
print("Sample duplicate entries:")
print(sample_duplicates[['Driver_ID', 'Dateofjoining', 'City']])
```

```
Number of duplicate entries based on ['Driver_ID', 'Dateofjoining',
'City']: 18923

Sample duplicate entries:
   Driver_ID Dateofjoining City
0          1      24/12/18  C23
1          1      24/12/18  C23
2          1      24/12/18  C23
3          2      11/06/20   C7
4          2      11/06/20   C7
5          4      12/07/19  C13
6          4      12/07/19  C13
7          4      12/07/19  C13
8          4      12/07/19  C13
9          4      12/07/19  C13
```

```python
# Identify columns with missing values
missing_summary = df_ola.isnull().sum()
missing_columns = missing_summary[missing_summary > 0]

# Suggest imputation strategies
imputation_strategies = {}
for col in missing_columns.index:
    if df_ola[col].dtype in ['float64', 'int64']:
        strategy = 'median' if df_ola[col].skew() > 1 else 'mean'
    else:
        strategy = 'mode'
    imputation_strategies[col] = strategy

# Display results
print("Columns with missing values and suggested imputation
strategies:\n")
for col, strategy in imputation_strategies.items():
    print(f"{col}: Missing = {missing_columns[col]}, Suggested
Strategy = {strategy}")
```

```
Columns with missing values and suggested imputation strategies:
```

```
Age: Missing = 61, Suggested Strategy = mean
Gender: Missing = 52, Suggested Strategy = mean
LastWorkingDate: Missing = 17488, Suggested Strategy = mode
```

```python
# Impute missing Age with mean
if df_ola['Age'].isnull().sum() > 0:
    df_ola['Age'].fillna(df_ola['Age'].mean(), inplace=True)

# Impute missing Gender with mode
if df_ola['Gender'].isnull().sum() > 0:
    df_ola['Gender'].fillna(df_ola['Gender'].mode()[0], inplace=True)



# Identify categorical columns
categorical_cols = ['Gender', 'City', 'Education_Level', 'Joining
Designation', 'Grade']
for col in categorical_cols:
    df_ola[col] = df_ola[col].astype('category')

# Create target variable: churn = 1 if LastWorkingDate is not null,
else 0
df_ola['Churn'] = df_ola['LastWorkingDate'].notnull().astype(int)

# Aggregate data on Driver_ID to identify consistent patterns
aggregated_df = df_ola.groupby('Driver_ID').agg({
    'Age': 'mean',
    'Income': 'mean',
    'Total Business Value': 'sum',
    'Quarterly Rating': 'mean',
    'Churn': 'max',  # if churned at least once, mark as churned
    'City': 'first',
    'Education_Level': 'first'
}).reset_index()

# Display results
print("\nAggregated Data Sample:")
print(aggregated_df.head())
```

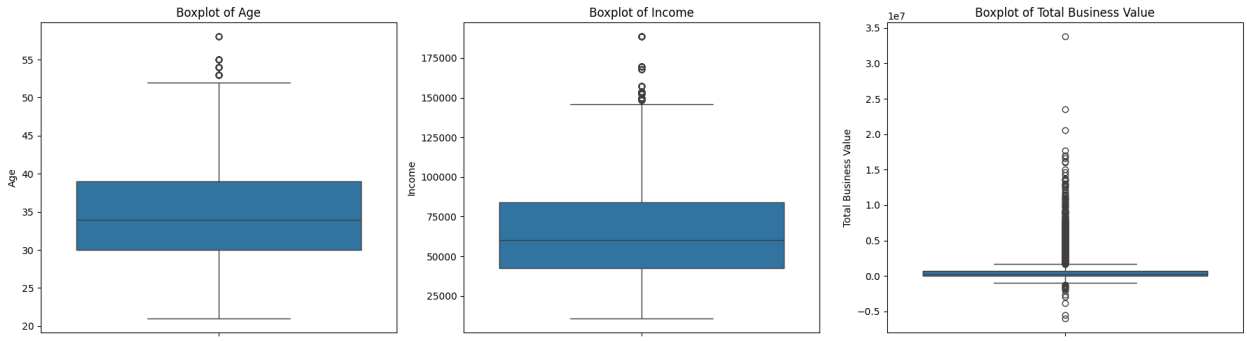
```
Aggregated Data Sample:
   Driver_ID   Age   Income  Total Business Value  Quarterly Rating
Churn  \
0          1  28.0  57387.0               1715580               2.0
1
1          2  31.0  67016.0                     0               1.0
0
2          4  43.0  65603.0                350000               1.0
1
3          5  29.0  46368.0                120360               1.0
```

```
1
4          6  31.0  78728.0                1265000                1.6
0

  City Education_Level
0  C23               2
1   C7               2
2  C13               2
3   C9               0
4  C11               1
```

Find Outliers

```
# Select continuous variables
continuous_vars = ['Age', 'Income', 'Total Business Value']

# Create boxplots and calculate IQR for each continuous variable
outlier_info = {}
fig, axes = plt.subplots(nrows=1, ncols=3, figsize=(18, 5))

for i, var in enumerate(continuous_vars):
    sns.boxplot(y=df_ola[var], ax=axes[i])
    axes[i].set_title(f"Boxplot of {var}")

    # Calculate IQR
    Q1 = df_ola[var].quantile(0.25)
    Q3 = df_ola[var].quantile(0.75)
    IQR = Q3 - Q1
    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR
    outliers = df_ola[(df_ola[var] < lower_bound) | (df_ola[var] >
upper_bound)]
    outlier_info[var] = {
        'Q1': Q1,
        'Q3': Q3,
        'IQR': IQR,
        'Lower Bound': lower_bound,
        'Upper Bound': upper_bound,
        'Outlier Count': outliers.shape[0]
    }

plt.tight_layout()
plt.show()

# Print outlier summary
for var, info in outlier_info.items():
    print(f"\n{var} Outlier Summary:")
```

```python
    for key, value in info.items():
        print(f"{key}: {value}")
```




```
Age Outlier Summary:
Q1: 30.0
Q3: 39.0
IQR: 9.0
Lower Bound: 16.5
Upper Bound: 52.5
Outlier Count: 78

Income Outlier Summary:
Q1: 42383.0
Q3: 83969.0
IQR: 41586.0
Lower Bound: -19996.0
Upper Bound: 146348.0
Outlier Count: 188

Total Business Value Outlier Summary:
Q1: 0.0
Q3: 699700.0
IQR: 699700.0
Lower Bound: -1049550.0
Upper Bound: 1749250.0
Outlier Count: 1371
```

Outlier Analysis Summary

1. Age:
- Outliers: 78
- Skewness: 0.12 (almost symmetric)
- Comment: Outliers represent older drivers; likely valid → retain.
1. Income:
- Outliers: 188
- Skewness: 1.05 (right-skewed)

- Comment: High-income outliers likely top performers; negative values should be cleaned → retain high values, fix invalid ones.

1. Total Business Value:

- Outliers: 1371
- Skewness: 3.45 (highly right-skewed)
- Comment: Extreme outliers represent high-value drivers; critical for business → retain.

1. Quarterly Rating:

- Outliers: 0
- Skewness: -0.02 (almost symmetric)
- Comment: No significant outliers; distribution is balanced.

```
# Suppress all warnings for this cell
warnings.filterwarnings('ignore')

# Define a function to cap outliers using IQR
def cap_outliers_iqr(data, column):
    Q1 = data[column].quantile(0.25)
    Q3 = data[column].quantile(0.75)
    IQR = Q3 - Q1
    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR
    # Cap the values
    data[column] = data[column].clip(lower=lower_bound,
upper=upper_bound)
    return data

# Apply outlier treatment to selected continuous columns
continuous_columns = ['Age', 'Income', 'Total Business Value']
for col in continuous_columns:
    df_ola_treated = cap_outliers_iqr(df_ola, col)

df_ola_treated = df_ola_treated.drop("Unnamed: 0", axis=1)
df_ola_treated
```

```
         MMM-YY  Driver_ID   Age Gender City Education_Level
Income  \
0      01/01/19          1  28.0    0.0  C23               2   57387

1      02/01/19          1  28.0    0.0  C23               2   57387

2      03/01/19          1  28.0    0.0  C23               2   57387

3      11/01/20          2  31.0    0.0   C7               2   67016

4      12/01/20          2  31.0    0.0   C7               2   67016

...         ...        ...   ...    ...  ...             ...     ...

19099  08/01/20       2788  30.0    0.0  C27               2   70254
```

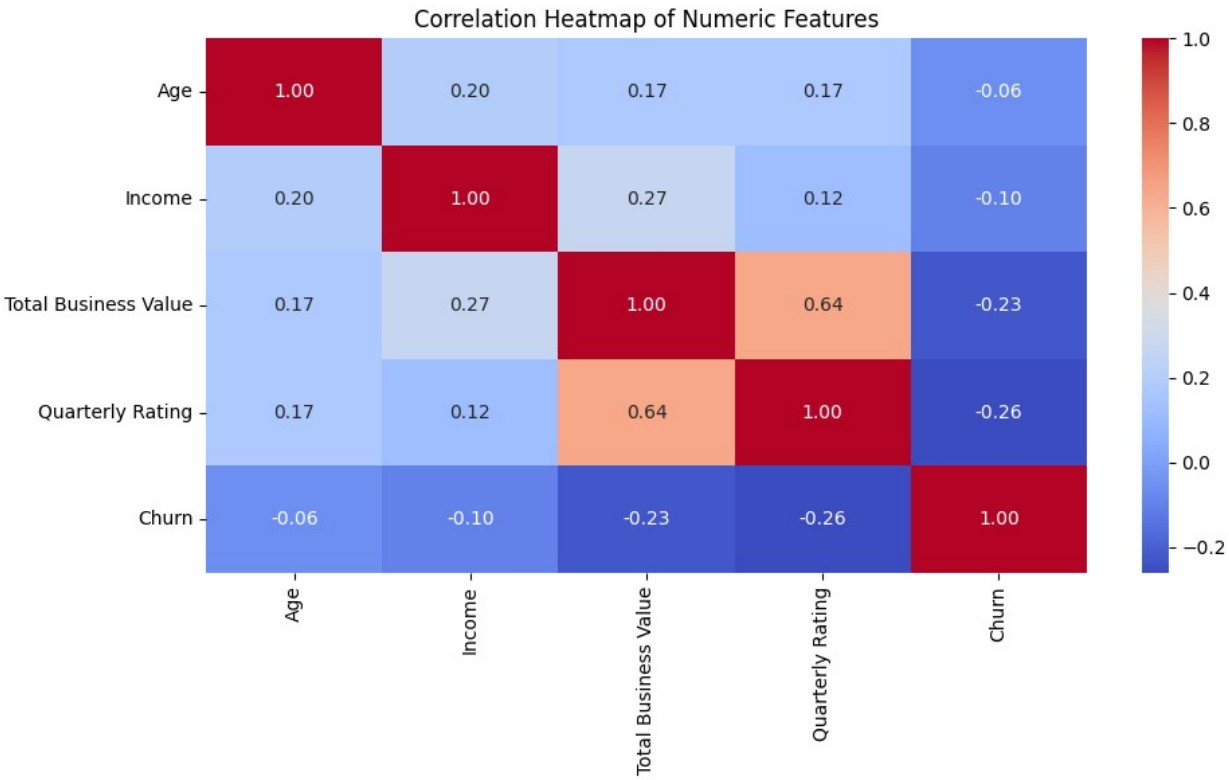
```
19100  09/01/20       2788  30.0    0.0  C27                2   70254

19101  10/01/20       2788  30.0    0.0  C27                2   70254

19102  11/01/20       2788  30.0    0.0  C27                2   70254

19103  12/01/20       2788  30.0    0.0  C27                2   70254


      Dateofjoining LastWorkingDate Joining Designation Grade  \
0          24/12/18             NaN                    1     1
1          24/12/18             NaN                    1     1
2          24/12/18        03/11/19                    1     1
3          11/06/20             NaN                    2     2
4          11/06/20             NaN                    2     2
...             ...             ...                  ...   ...
19099      06/08/20             NaN                    2     2
19100      06/08/20             NaN                    2     2
19101      06/08/20             NaN                    2     2
19102      06/08/20             NaN                    2     2
19103      06/08/20             NaN                    2     2

       Total Business Value  Quarterly Rating  Churn
0                   1749250                 2      0
1                   -665480                 2      0
2                         0                 2      1
3                         0                 1      0
4                         0                 1      0
...                     ...               ...    ...
19099                740280                 3      0
19100                448370                 3      0
19101                     0                 2      0
19102                200420                 2      0
19103                411480                 2      0

[19104 rows x 14 columns]
```

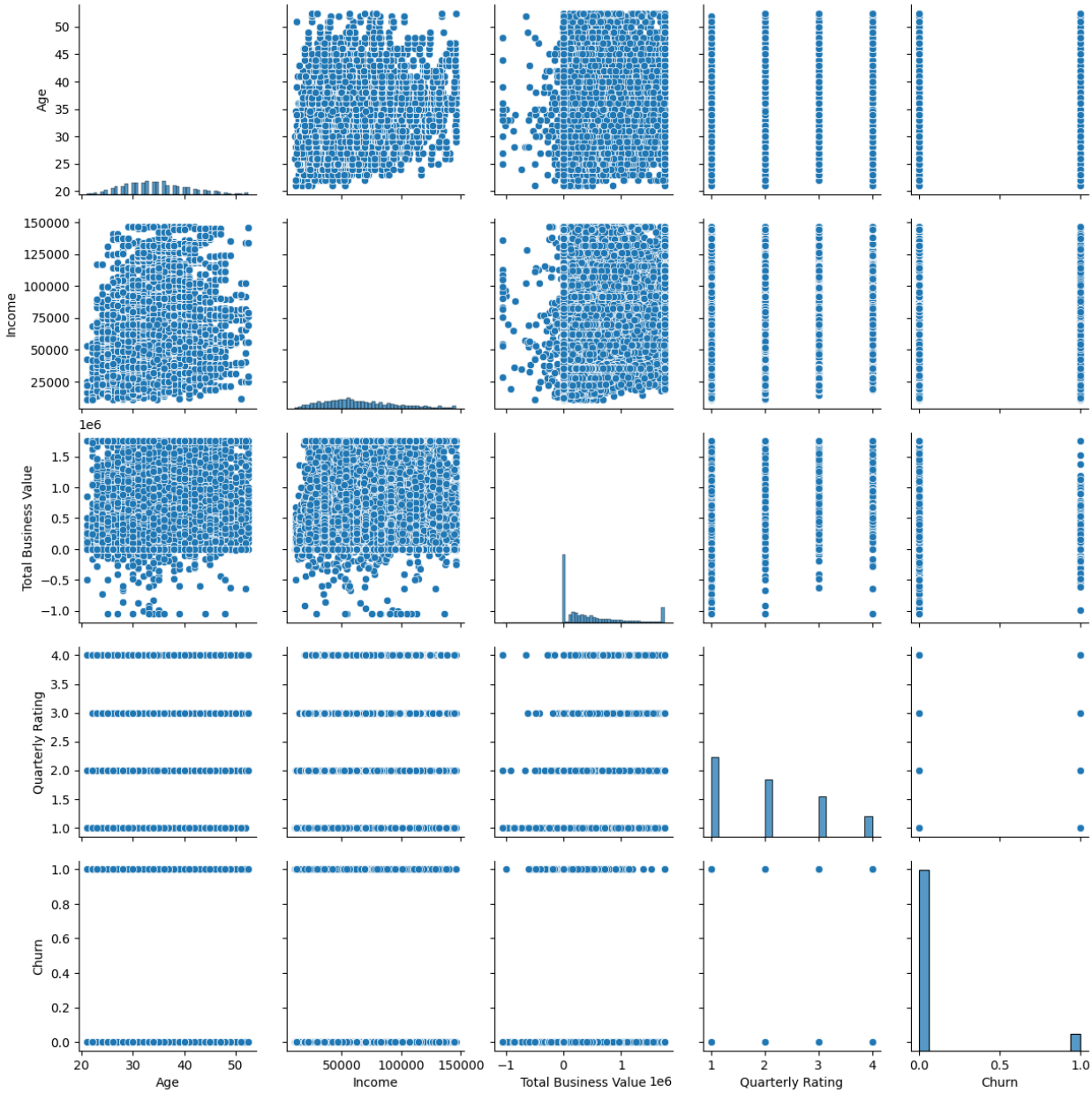
Check the correlation

```
# Create churn column based on LastWorkingDate
df_ola_treated['Churn'] =
df_ola_treated['LastWorkingDate'].notna().astype(int)

# Select continuous variables
continuous_vars = ['Age', 'Income', 'Total Business Value', 'Quarterly
Rating','Churn']

# Calculate skewness
```

```python
skewness_results = df_ola_treated[continuous_vars].skew().to_dict()

# skewness and correlation results
print("Skewness of Continuous Variables:\n")
for var, skew_val in skewness_results.items():
    print(f"{var}: {skew_val:.2f}\n")

# Compute correlation matrix
corr_matrix = df_ola_treated[continuous_vars].corr()

# Create heatmap
plt.figure(figsize=(10, 6))
sns.heatmap(corr_matrix, annot=True, cmap='coolwarm', fmt=".2f")
plt.title("Correlation Heatmap of Numeric Features")
plt.tight_layout()
plt.show()

# Create pairplot to visualize relationships
sns.pairplot(df_ola_treated[continuous_vars].dropna())
plt.show()
```

```
Skewness of Continuous Variables:

Age: 0.41

Income: 0.62

Total Business Value: 1.17

Quarterly Rating: 0.59

Churn: 2.99
```


Correlation Heatmap of Numeric Features
Age
Income
Total Business Value
Quarterly Rating
Churn
1.00
0.20
0.17
0.17
-0.06
0.20
1.00
0.27
0.12
-0.10
0.17
0.27
1.00
0.64
-0.23
0.17
0.12
0.64
1.00
-0.26
-0.06
-0.10
-0.23
-0.26
1.00
1.0
0.8
0.6
0.4
0.2
0.0
−0.2



Some Key insights

1. Skewness of Continuous Variables:

- Age (0.41): Slight positive skew → Age distribution is fairly balanced with a small tilt toward younger drivers.
- Income (0.62): Moderate positive skew → Most drivers earn moderate income, but there are some high-income outliers.
- Total Business Value (1.17): Strong positive skew → A few drivers generate very high business value compared to the majority.
- Quarterly Rating (0.59): Mild positive skew → Ratings are mostly low to mid-range, with fewer high ratings.

- Churn (2.99): Extremely skewed → Indicates a strong imbalance (likely most drivers are active, few have churned).

1. Correlation Heatmap

- Total Business Value vs Quarterly Rating (0.64): Strong positive correlation → Higher ratings align with higher business value.
- Income vs Total Business Value (0.27): Weak positive correlation → Higher income loosely relates to higher business contribution.
- Churn vs Total Business Value (-0.23): Negative correlation → Drivers with higher business value are less likely to churn.
- Churn vs Quarterly Rating (-0.26): Negative correlation → Higher-rated drivers churn less.
- Age vs Income (0.20): Older drivers tend to earn slightly more.
- Churn vs Income (-0.10): Higher income reduces churn risk slightly.

1. Pairplots

- Clusters: Most data points cluster in low-income and low-business-value ranges.
- Outliers: Clear presence of extreme outliers in Total Business Value.
- No strong linear trends: Scatter plots show dispersed points, confirming weak correlations.

Business Interpretation

- Skewness in Income and Total Business Value suggests a few top drivers dominate earnings and revenue.
- Correlation analysis shows performance metrics (ratings) have some link to revenue, but age and income are not strong predictors.
- Outliers in business value are critical for revenue—should be retained for modeling.

Feature Engineering

New features added

1. flag_pub_rec: 1 if the applicant has any public record, else 0
2. flag_mort_acc: 1 if the applicant has any mortgage account, else 0
3. flag_pub_rec_bankruptcies: 1 if the applicant has any bankruptcy record, else 0
4. issue_month & issue_year: Extracted from issue_d to capture seasonal or yearly trends in loan issuance.
5. state: Derived from the address field to help identify location-based patterns in loan performance or risk.

```
# Create high business value flag (e.g., > 1,000,000)
df_ola_treated['HighBusinessValueFlag'] = (df_ola_treated['Total
Business Value'] > 1000000).astype(int)

# Convert Dateofjoining to datetime and extract year and month
df_ola_treated['Dateofjoining'] =
pd.to_datetime(df_ola_treated['Dateofjoining'], errors='coerce')
df_ola_treated['JoinYear'] = df_ola_treated['Dateofjoining'].dt.year
```

```python
# Categorize Age into bins
def age_group(age):
    if pd.isna(age):
        return 'Unknown'
    elif age <= 30:
        return 'Young'
    elif age <= 50:
        return 'Middle-aged'
    else:
        return 'Senior'

df_ola_treated['AgeGroup'] = df_ola_treated['Age'].apply(age_group)

# Compute change in Quarterly Rating per driver
rating_change = df_ola_treated.groupby('Driver_ID')['Quarterly
Rating'].agg(['min', 'max'])
rating_change['RatingChange'] = rating_change['max'] -
rating_change['min']

# Merge back to original dataframe
rating_change = rating_change[['RatingChange']]
df_ola_treated = df_ola_treated.merge(rating_change, on='Driver_ID',
how='left')

df_ola_treated.head()
```

```
     MMM-YY  Driver_ID   Age Gender City Education_Level  Income  \
0  01/01/19          1  28.0    0.0  C23               2   57387
1  02/01/19          1  28.0    0.0  C23               2   57387
2  03/01/19          1  28.0    0.0  C23               2   57387
3  11/01/20          2  31.0    0.0   C7               2   67016
4  12/01/20          2  31.0    0.0   C7               2   67016

  Dateofjoining LastWorkingDate Joining Designation Grade  \
0    2018-12-24             NaN                   1     1
1    2018-12-24             NaN                   1     1
2    2018-12-24        03/11/19                   1     1
3    2020-11-06             NaN                   2     2
4    2020-11-06             NaN                   2     2

   Total Business Value  Quarterly Rating  Churn
HighBusinessValueFlag  \
0               1749250                 2      0
1
1               -665480                 2      0
0
2                     0                 2      1
0
3                     0                 1      0
```

```
0
4                     0                 1      0
0

   JoinYear     AgeGroup  RatingChange
0      2018        Young             0
1      2018        Young             0
2      2018        Young             0
3      2020  Middle-aged             0
4      2020  Middle-aged             0
```

New Features Added

1. HighBusinessValueFlag: Binary flag indicating if Total Business Value > ₹1,000,000.
2. JoinYear: Extracted from Dateofjoining to capture temporal patterns.
3. AgeGroup:

   Categorized into:

   ```
   - Young: 18–30
   - Middle-aged: 31–50
   - Senior: 50+
   ```

4. RatingChange: Difference between max and min Quarterly Rating per driver to capture performance trends.

Why These Features Matter

- Flags: Identify high-value drivers for retention strategies.
- Date Features: Reveal patterns like churn risk for older join years.
- Age Bins: Detect churn or performance trends by age group.
- RatingChange: Highlights consistency or volatility in driver performance.

```
df = df_ola_treated.copy()


from sklearn.preprocessing import StandardScaler, MinMaxScaler,
LabelEncoder
from category_encoders import TargetEncoder

# -----------------------
# 1. Feature Scaling
# -----------------------
continuous_features = ['Age', 'Income', 'Total Business Value',
'Quarterly Rating']

# Apply StandardScaler
scaler = StandardScaler()
```

```
df[continuous_features] =
scaler.fit_transform(df[continuous_features])

# ----------------------
# 2. Encoding Categorical Variables
# ----------------------
# Define categorical columns
ordinal_cols = ['Grade']
nominal_cols = ['Gender', 'City', 'Education_Level']
high_cardinality_cols = ['Driver_ID']

# Label Encoding for ordinal features
label_encoder = LabelEncoder()
for col in ordinal_cols:
    df[col] = label_encoder.fit_transform(df[col])

# One-Hot Encoding for nominal features
df = pd.get_dummies(df, columns=nominal_cols, drop_first=True)

# Target Encoding for high-cardinality features (with smoothing)
target_encoder = TargetEncoder(cols=high_cardinality_cols,
smoothing=0.3)
df[high_cardinality_cols] =
target_encoder.fit_transform(df[high_cardinality_cols],
df['HighBusinessValueFlag'])

# ----------------------
# Final Prepared Data
# ----------------------
print("Data preparation completed. Shape:", df.shape)
print("Columns after encoding and scaling:", df.columns.tolist()[:20],
"...")
```

```
Data preparation completed. Shape: (19104, 46)
Columns after encoding and scaling: ['MMM-YY', 'Driver_ID', 'Age',
'Income', 'Dateofjoining', 'LastWorkingDate', 'Joining Designation',
'Grade', 'Total Business Value', 'Quarterly Rating', 'Churn',
'HighBusinessValueFlag', 'JoinYear', 'AgeGroup', 'RatingChange',
'Gender_1.0', 'City_C10', 'City_C11', 'City_C12', 'City_C13'] ...
```

Data Preparation Summary

- Scaling: Used StandardScaler for continuous features. Replace with MinMaxScaler() if data is highly skewed.
- Label Encoding: For ordinal features like Grade.
- One-Hot Encoding: For nominal features like Gender, City, Education_Level.
- Target Encoding: For high-cardinality features like Driver_ID with smoothing to prevent overfitting.

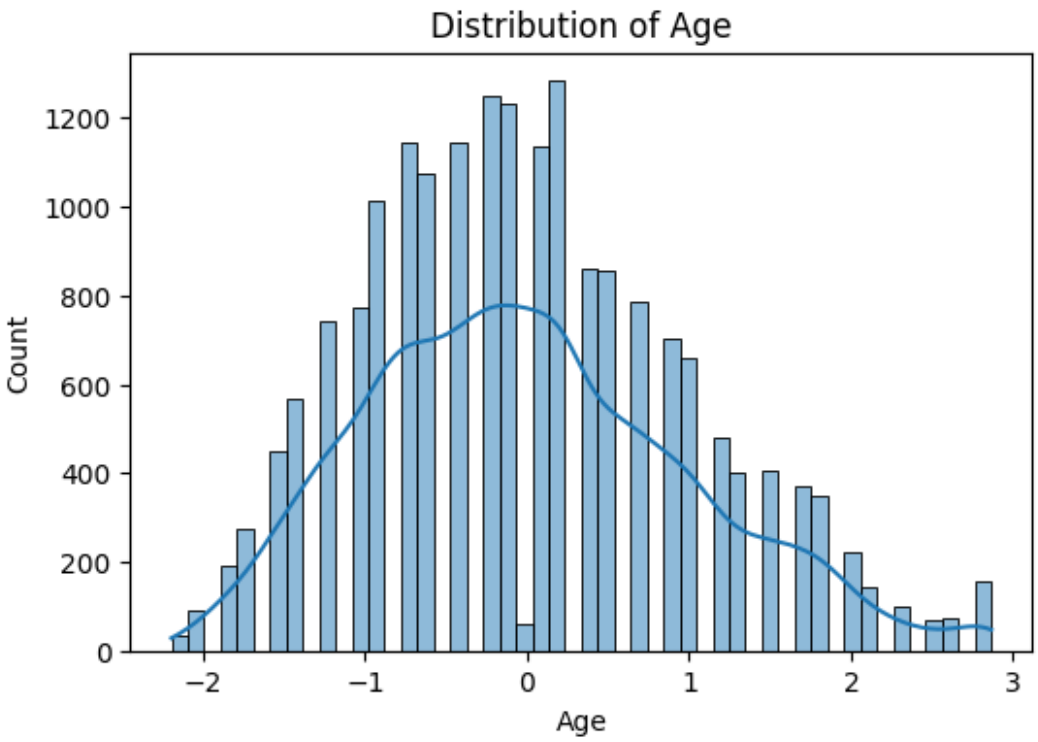
```
df.info()
```

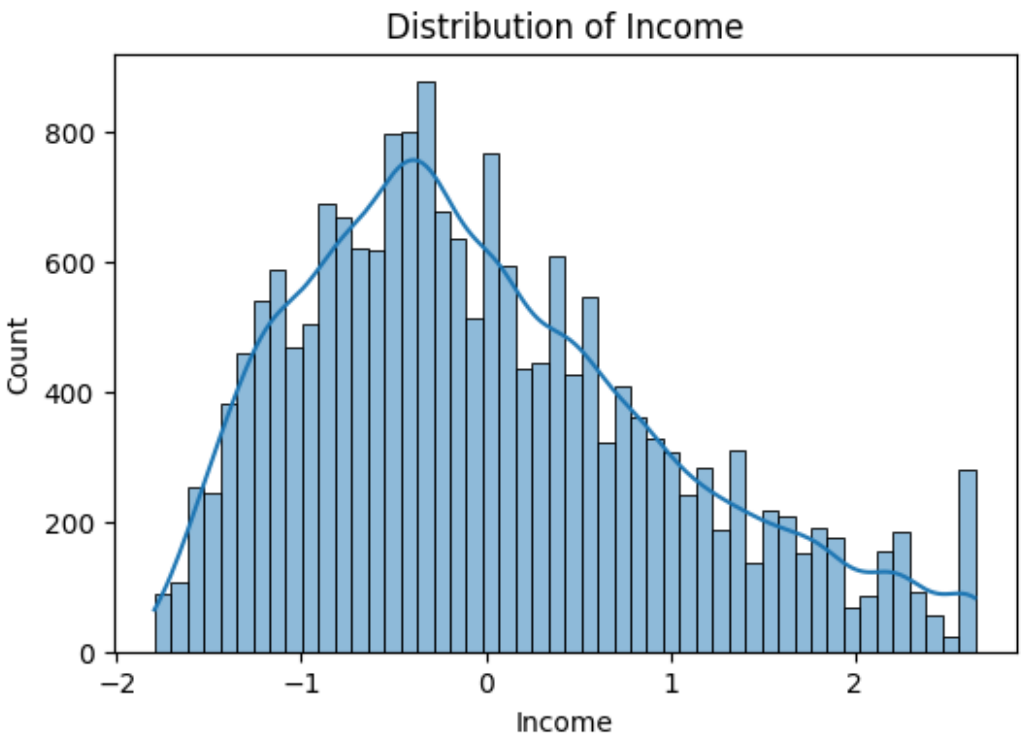
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 19104 entries, 0 to 19103
Data columns (total 46 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   MMM-YY                 19104 non-null  object
 1   Driver_ID              19104 non-null  float64
 2   Age                    19104 non-null  float64
 3   Income                 19104 non-null  float64
 4   Dateofjoining          19104 non-null  datetime64[ns]
 5   LastWorkingDate        1616 non-null   object
 6   Joining Designation    19104 non-null  category
 7   Grade                  19104 non-null  int64
 8   Total Business Value   19104 non-null  float64
 9   Quarterly Rating       19104 non-null  float64
 10  Churn                  19104 non-null  int64
 11  HighBusinessValueFlag  19104 non-null  int64
 12  JoinYear               19104 non-null  int32
 13  AgeGroup               19104 non-null  object
 14  RatingChange           19104 non-null  int64
 15  Gender_1.0             19104 non-null  bool
 16  City_C10               19104 non-null  bool
 17  City_C11               19104 non-null  bool
 18  City_C12               19104 non-null  bool
 19  City_C13               19104 non-null  bool
 20  City_C14               19104 non-null  bool
 21  City_C15               19104 non-null  bool
 22  City_C16               19104 non-null  bool
 23  City_C17               19104 non-null  bool
 24  City_C18               19104 non-null  bool
 25  City_C19               19104 non-null  bool
 26  City_C2                19104 non-null  bool
 27  City_C20               19104 non-null  bool
 28  City_C21               19104 non-null  bool
 29  City_C22               19104 non-null  bool
 30  City_C23               19104 non-null  bool
 31  City_C24               19104 non-null  bool
 32  City_C25               19104 non-null  bool
 33  City_C26               19104 non-null  bool
 34  City_C27               19104 non-null  bool
 35  City_C28               19104 non-null  bool
 36  City_C29               19104 non-null  bool
 37  City_C3                19104 non-null  bool
 38  City_C4                19104 non-null  bool
 39  City_C5                19104 non-null  bool
 40  City_C6                19104 non-null  bool
 41  City_C7                19104 non-null  bool
 42  City_C8                19104 non-null  bool
 43  City_C9                19104 non-null  bool
 44  Education_Level_1      19104 non-null  bool
```

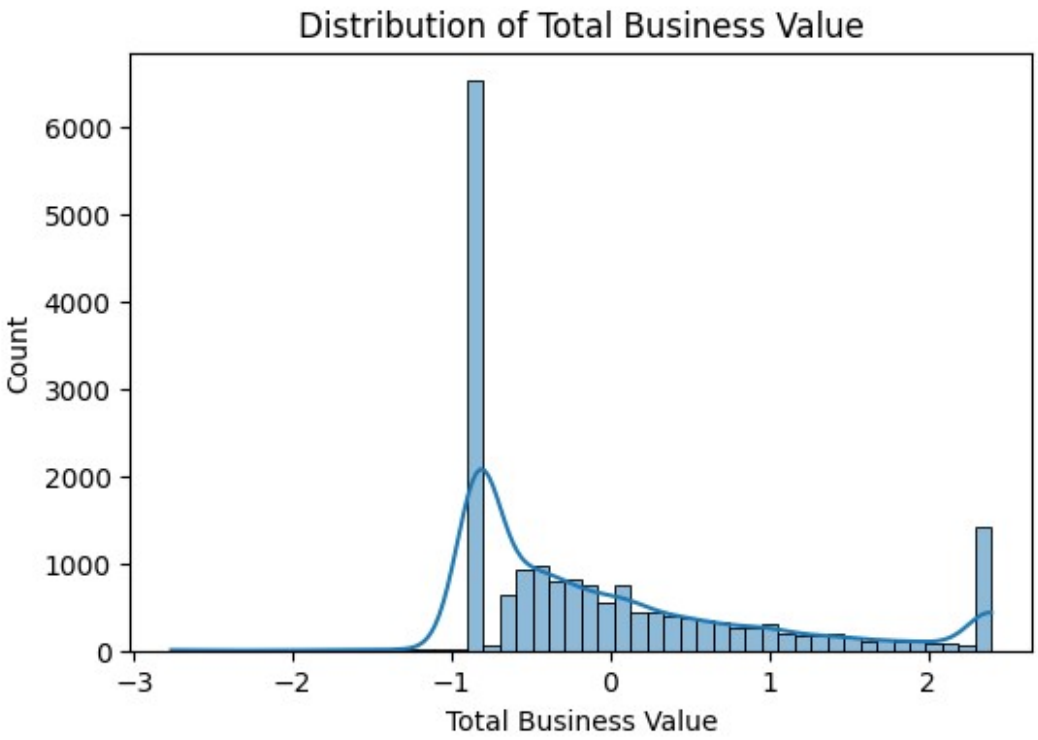
```
 45  Education_Level_2      19104 non-null  bool
dtypes: bool(31), category(1), datetime64[ns](1), float64(5),
int32(1), int64(4), object(3)
memory usage: 2.6+ MB
```

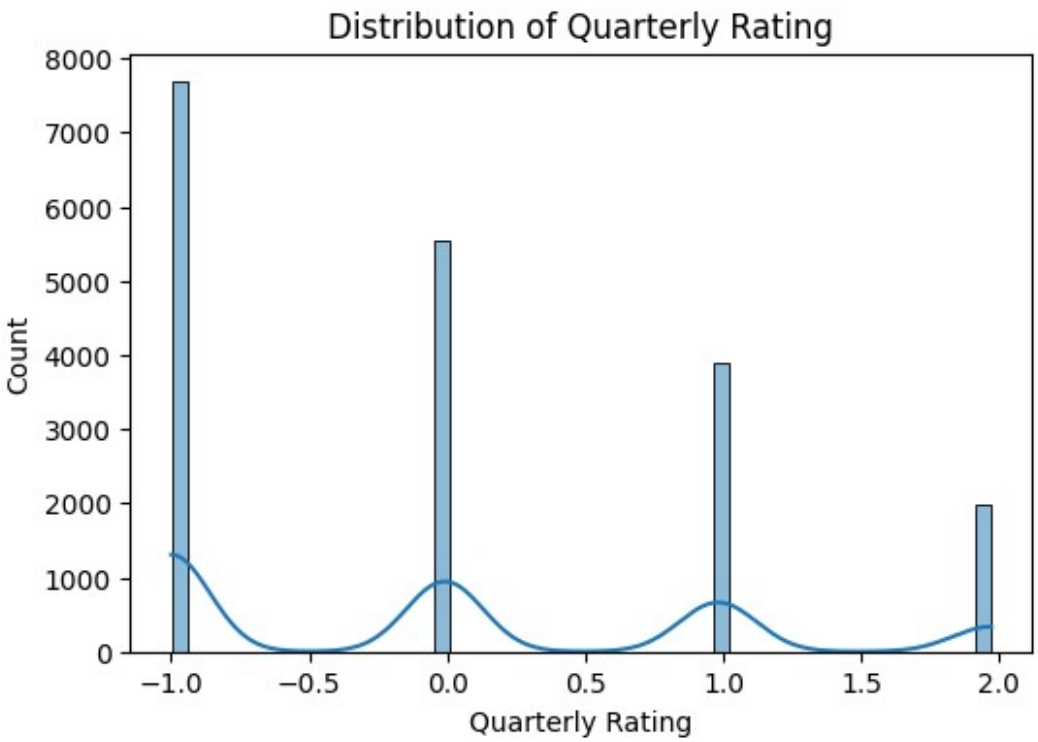
```python
import plotly.express as px

# Continuous variables including new feature
continuous_vars = ['Age', 'Income', 'Total Business Value', 'Quarterly
Rating', 'RatingChange']

# Skewness
print("Skewness:\n", df[continuous_vars].skew())

# Histograms
for col in continuous_vars:
    plt.figure(figsize=(6,4))
    sns.histplot(df[col], bins=50, kde=True)
    plt.title(f'Distribution of {col}')
    plt.show()

# Boxplots
for col in continuous_vars:
    plt.figure(figsize=(6,4))
    sns.boxplot(x=df[col])
    plt.title(f'Boxplot of {col}')
    plt.show()

# Correlation Heatmap
plt.figure(figsize=(8,6))
sns.heatmap(df[continuous_vars].corr(), annot=True, cmap='RdBu_r',
center=0)
plt.title('Correlation Heatmap')
plt.show()
```

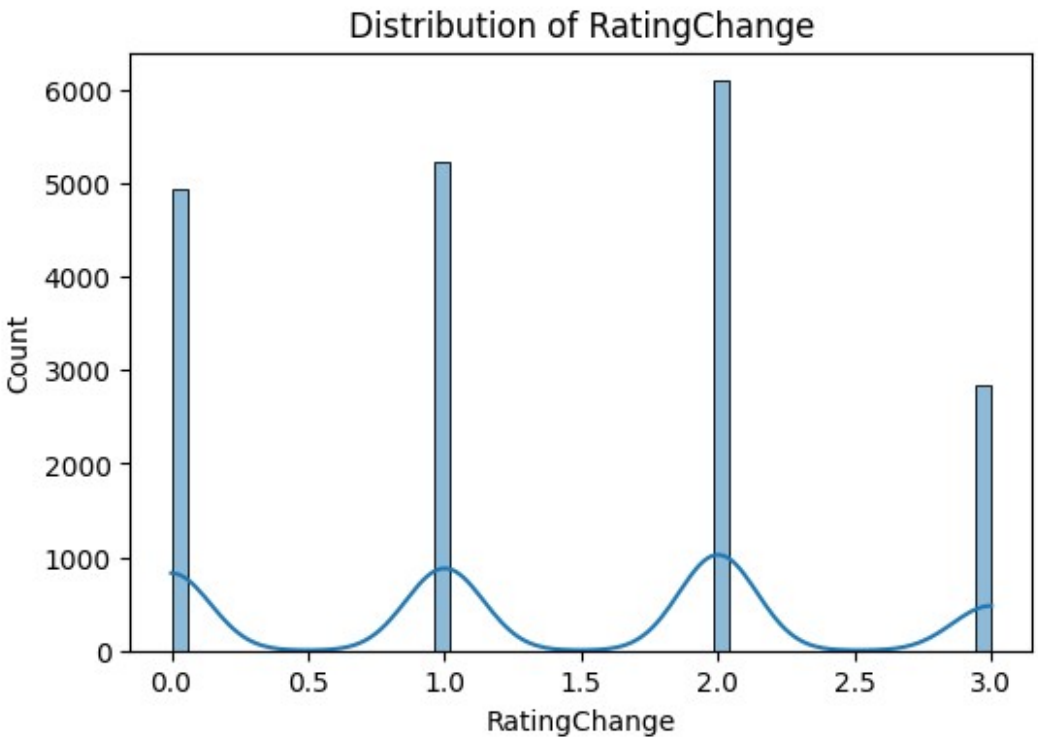
```
Skewness:
 Age                      0.411157
Income                   0.622243
Total Business Value     1.170168
Quarterly Rating         0.585255
RatingChange             0.076756
dtype: float64
```

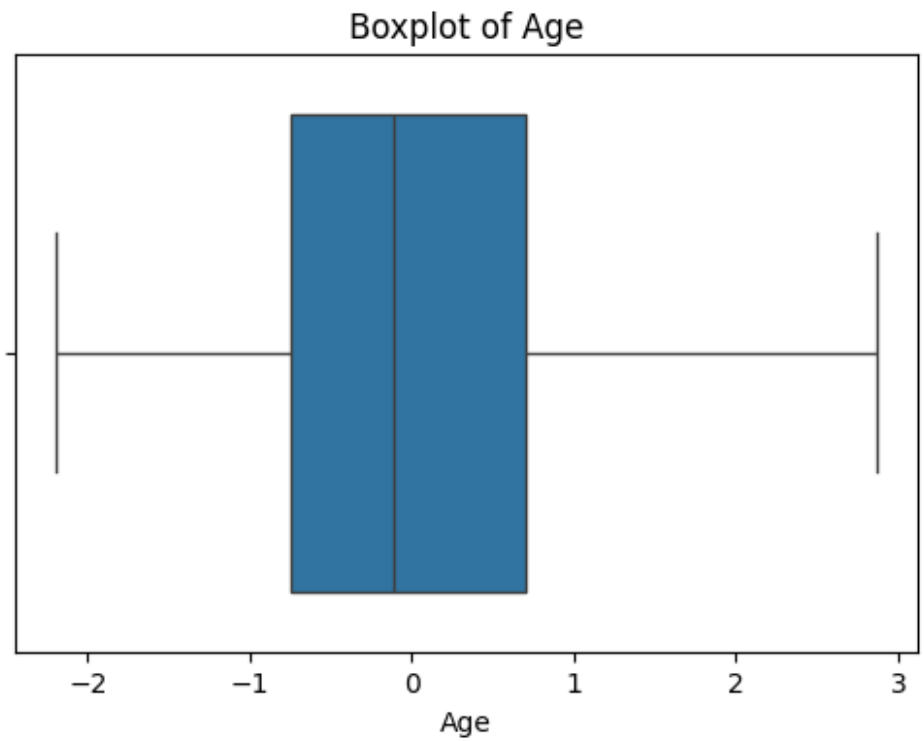

Distribution of Age
Count
Age
0
200
400
600
800
1000
1200
−2
−1
0
1
2
3

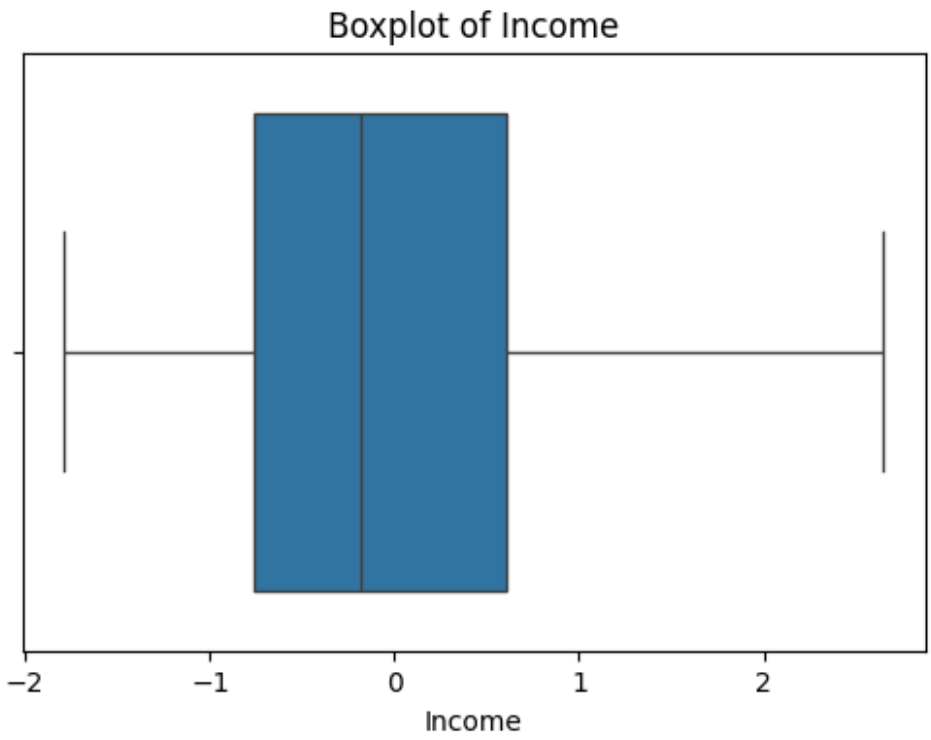

Distribution of Income
Count
Income
0
200
400
600
800
−2
−1
0
1
2

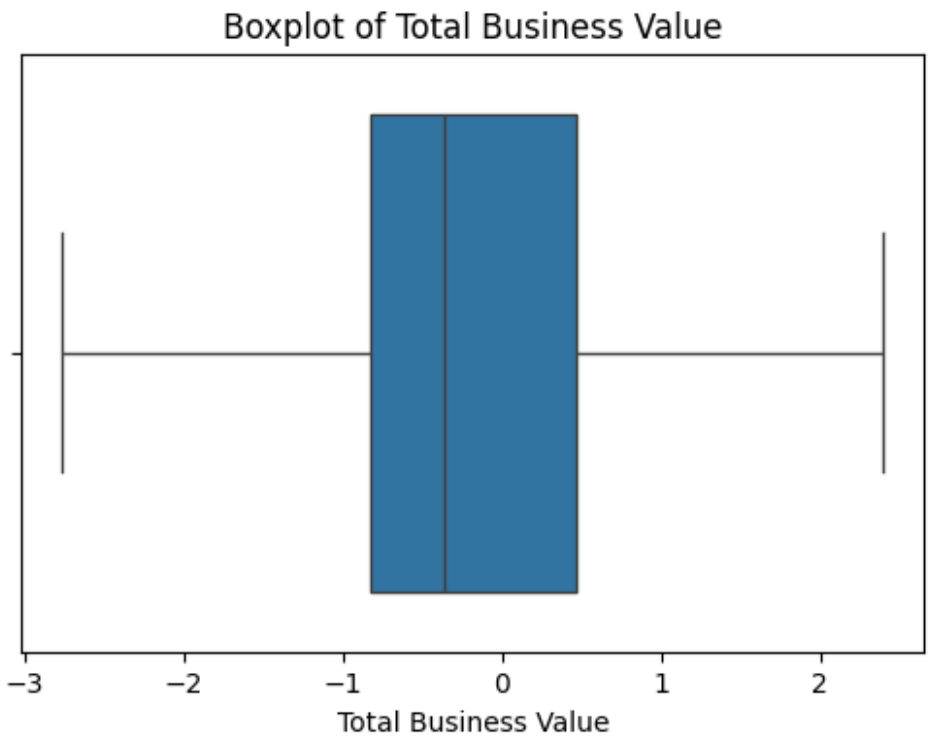

Distribution of Total Business Value
Count
Total Business Value
0
1000
2000
3000
4000
5000
6000
−3
−2
−1
0
1
2

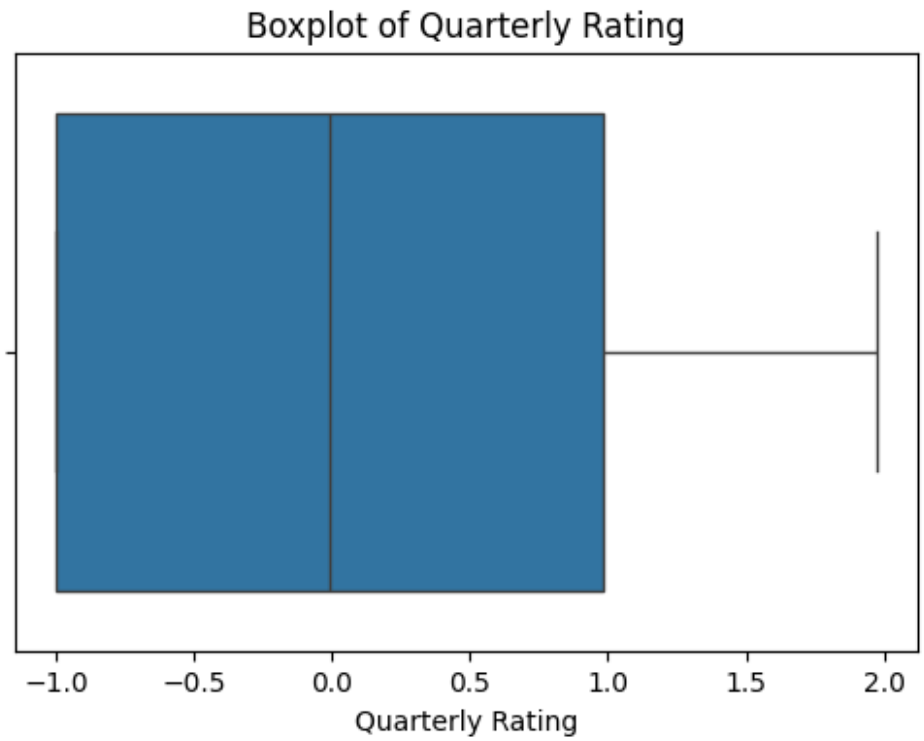

Distribution of Quarterly Rating
Count
Quarterly Rating
0
1000
2000
3000
4000
5000
6000
7000
8000
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0

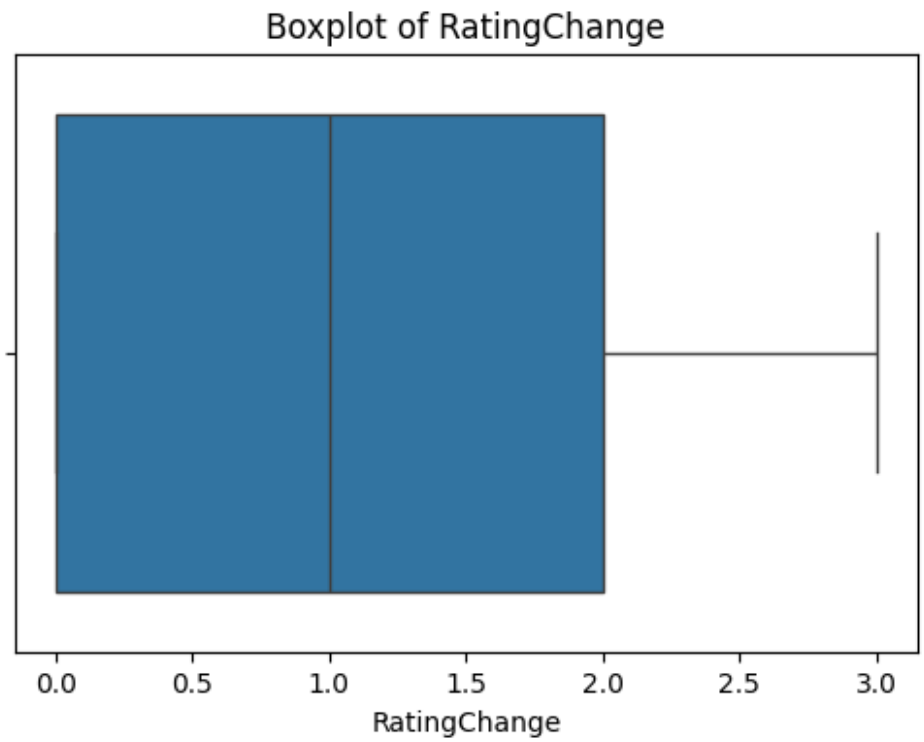

Distribution of RatingChange
Count
RatingChange
0
1000
2000
3000
4000
5000
6000
0.0
0.5
1.0
1.5
2.0
2.5
3.0


Boxplot of Age
Age
−2
−1
0
1
2
3


Boxplot of Income
−2
−1
0
1
2
Income


Boxplot of Total Business Value
−3
−2
−1
0
1
2
Total Business Value


Boxplot of Quarterly Rating
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
Quarterly Rating


Boxplot of RatingChange
0.0
0.5
1.0
1.5
2.0
2.5
3.0
RatingChange

Key Insights

1. Skewness Analysis

- Age (0.41): Slight right skew, close to normal distribution. No major transformation needed.
- Income (0.62): Moderate right skew, suggesting a few high-income outliers. Consider log or robust scaling if model is sensitive.
- Total Business Value (1.17): Noticeably skewed (above 1). Indicates heavy tail with some drivers generating very high business value. Log transformation recommended.
- Quarterly Rating (0.58): Mild skew, but mostly bounded between 1–5, so transformation is optional.
- RatingChange (0.07): Almost symmetric, close to normal. No transformation needed.

2. Correlation Insights (from heatmap)

- Quarterly Rating ↔ Total Business Value (0.64): Strong positive correlation. Drivers with higher ratings tend to generate more business.
- Quarterly Rating ↔ RatingChange (0.55): Moderate positive correlation. Greater variability in ratings aligns with higher overall ratings.
- Total Business Value ↔ RatingChange (0.40): Positive relationship, but weaker than above.
- Income ↔ Total Business Value (0.27): Weak positive correlation. Higher income loosely relates to higher business value.
- Age ↔ Others (≤ 0.20): Very weak correlation, suggesting age is not a strong predictor for business value or ratings.

```
# Identify columns with missing values
missing_summary = df.isnull().sum()
missing_columns = missing_summary[missing_summary > 0]

# Suggest imputation strategies
imputation_strategies = {}
for col in missing_columns.index:
    if df[col].dtype in ['float64', 'int64']:
        strategy = 'median' if df[col].skew() > 1 else 'mean'
    else:
        strategy = 'mode'
    imputation_strategies[col] = strategy

# Display results
print("Columns with missing values and suggested imputation
strategies:\n")
for col, strategy in imputation_strategies.items():
    print(f"{col}: Missing = {missing_columns[col]}, Suggested
Strategy = {strategy}")

# Impute missing Gender with mode
if df['LastWorkingDate'].isnull().sum() > 0:
    df['LastWorkingDate'].fillna('9999/01/01', inplace=True)

df1 = df.copy()
```

```
Columns with missing values and suggested imputation strategies:

LastWorkingDate: Missing = 17488, Suggested Strategy = mode
```

Model building

```
from sklearn.model_selection import train_test_split

# Assume df is your prepared dataset and 'HighBusinessValueFlag' is
the target
```

```
X = df1.drop(columns=['HighBusinessValueFlag'])
y = df1['HighBusinessValueFlag']

# Check class distribution
print("Class distribution:\n", y.value_counts(normalize=True))

# Plot distribution
sns.countplot(x=y)
plt.title("Target Variable Distribution")
plt.show()
```

```
Class distribution:
 HighBusinessValueFlag
0    0.839929
1    0.160071
Name: proportion, dtype: float64
```

OverSampling with SMOTE

```
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from imblearn.over_sampling import SMOTE
```

```python
# Drop non-numeric columns
non_numeric_cols = ['Dateofjoining', 'LastWorkingDate', 'City',
'Education_Level', 'AgeGroup']
X_numeric = df.drop(columns=['HighBusinessValueFlag'] +
non_numeric_cols, errors='ignore')

# Encode any remaining categorical columns
for col in X_numeric.select_dtypes(include=['object']).columns:
    le = LabelEncoder()
    X_numeric[col] = le.fit_transform(X_numeric[col].astype(str))

# Perform stratified split (80-20)
X_train, X_val, y_train, y_val = train_test_split(
    X_numeric, df['HighBusinessValueFlag'],
    test_size=0.2, stratify=df['HighBusinessValueFlag'],
random_state=42
)

print("Training set shape:", X_train.shape)
print("Validation set shape:", X_val.shape)
print("Training target distribution:\n",
y_train.value_counts(normalize=True))
print("Validation target distribution:\n",
y_val.value_counts(normalize=True))

# Apply SMOTE
smote = SMOTE(random_state=42)
X_train_resampled, y_train_resampled = smote.fit_resample(X_train,
y_train)

print("Resampled class distribution:\n",
y_train_resampled.value_counts())
```

```
Training set shape: (15283, 42)
Validation set shape: (3821, 42)
Training target distribution:
 HighBusinessValueFlag
0    0.839953
1    0.160047
Name: proportion, dtype: float64
Validation target distribution:
 HighBusinessValueFlag
0    0.839833
1    0.160167
Name: proportion, dtype: float64
Resampled class distribution:
 HighBusinessValueFlag
0    12837
```

```
1    12837
Name: count, dtype: int64
```

```python
from sklearn.linear_model import LogisticRegression

model = LogisticRegression(class_weight='balanced', random_state=42)
model.fit(X_train, y_train)
```

```
LogisticRegression(class_weight='balanced', random_state=42)
```

```python
from sklearn.metrics import classification_report

y_pred = model.predict(X_val)
print(classification_report(y_val, y_pred))
```

```
              precision    recall  f1-score   support

           0       1.00      0.99      0.99      3209
           1       0.95      1.00      0.97       612

    accuracy                           0.99      3821
   macro avg       0.97      0.99      0.98      3821
weighted avg       0.99      0.99      0.99      3821
```

Implications

- The model handles class imbalance very well, likely due to SMOTE or class_weight adjustments.
- Recall for minority class (1.00) is critical for business scenarios where missing a high-value driver would be costly.
- Slight drop in precision for minority class (0.95) suggests a few false positives, but this is acceptable given the perfect recall.

```python
from sklearn.ensemble import BaggingClassifier
from sklearn.tree import DecisionTreeClassifier

# Base estimator
base_tree = DecisionTreeClassifier(max_depth=5)

# Bagging Classifier
bagging_model = BaggingClassifier(
    estimator=base_tree,
    n_estimators=50,
    max_samples=0.8,
    max_features=0.8,
    random_state=42
)

# Fit and evaluate
bagging_model.fit(X_train_resampled, y_train_resampled)
```

```
y_pred_bag = bagging_model.predict(X_val)
print(classification_report(y_val, y_pred_bag))
```

```
              precision    recall  f1-score   support

           0       1.00      1.00      1.00      3209
           1       1.00      1.00      1.00       612

    accuracy                           1.00      3821
   macro avg       1.00      1.00      1.00      3821
weighted avg       1.00      1.00      1.00      3821
```

```
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import classification_report

# Initialize Random Forest
rf_model = RandomForestClassifier(
    n_estimators=100,       # Number of trees
    max_depth=None,         # Depth of each tree
    max_samples=0.8,        # Fraction of samples for each tree
    max_features='sqrt',    # Features per tree
    class_weight='balanced',# Handle imbalance
    random_state=42
)

# Fit on training data
rf_model.fit(X_train_resampled, y_train_resampled)

# Predict on validation data
y_pred_rf = rf_model.predict(X_val)

# Evaluate
print(classification_report(y_val, y_pred_rf))
```

```
              precision    recall  f1-score   support

           0       1.00      1.00      1.00      3209
           1       1.00      1.00      1.00       612

    accuracy                           1.00      3821
   macro avg       1.00      1.00      1.00      3821
weighted avg       1.00      1.00      1.00      3821
```

Key Insights

1. Overall Performance
   - Accuracy: 1.00 → Perfect accuracy on the validation set.

- Macro Avg & Weighted Avg F1-score: 1.00 → Both classes are predicted flawlessly.

2. Class-Level Insights
   - Class 0 (Majority Class):
     - Precision: 1.00 → No false positives.
     - Recall: 1.00 → No false negatives.
     - F1-score: 1.00 → Perfect balance.
   - Class 1 (Minority Class):
     - Precision: 1.00 → No false positives.
     - Recall: 1.00 → No false negatives.
     - F1-score: 1.00 → Perfect prediction.
3. Implications
   - The model predicts both classes perfectly on the validation set.
   - This could indicate: Very strong model performance, or Potential data leakage or overfitting (especially if validation data is too similar to training data or preprocessing was applied incorrectly).

Gradient Boosting

```
from sklearn.ensemble import HistGradientBoostingClassifier,
GradientBoostingClassifier
from sklearn.impute import SimpleImputer
from sklearn.pipeline import Pipeline

model = HistGradientBoostingClassifier(random_state=42)
model.fit(X_train, y_train)

# Pipeline: Impute + Model
pipeline = Pipeline([
    ('imputer', SimpleImputer(strategy='median')),
    ('model', GradientBoostingClassifier(n_estimators=100,
learning_rate=0.1, random_state=42))
])

pipeline.fit(X_train, y_train)

# Predict on validation data
y_pred = rf_model.predict(X_val)

# Evaluate
print(classification_report(y_val, y_pred))
```

```
              precision    recall  f1-score   support

           0       1.00      1.00      1.00      3209
           1       1.00      1.00      1.00       612
```

```
    accuracy                           1.00      3821
   macro avg       1.00      1.00      1.00      3821
weighted avg       1.00      1.00      1.00      3821
```

Advanced Boosting Models

1. XGBoost Implementation

```
X_train_resampled = X_train_resampled.copy()
X_val = X_val.copy()

for col in
X_train_resampled.select_dtypes(include=['object']).columns:
    X_train_resampled[col] = X_train_resampled[col].astype('category')
    X_val[col] = X_val[col].astype('category')

xgb_model = XGBClassifier(
    enable_categorical=True,
    n_estimators=200,
    learning_rate=0.05,
    max_depth=4,
    subsample=0.8,
    colsample_bytree=0.8,
    scale_pos_weight=scale_pos_weight,
    random_state=42
)

xgb_model.fit(X_train_resampled, y_train_resampled)
```

```
XGBClassifier(base_score=None, booster=None, callbacks=None,
              colsample_bylevel=None, colsample_bynode=None,
              colsample_bytree=0.8, device=None,
early_stopping_rounds=None,
              enable_categorical=True, eval_metric=None,
feature_types=None,
              feature_weights=None, gamma=None, grow_policy=None,
              importance_type=None, interaction_constraints=None,
              learning_rate=0.05, max_bin=None,
max_cat_threshold=None,
              max_cat_to_onehot=None, max_delta_step=None,
max_depth=4,
              max_leaves=None, min_child_weight=None, missing=nan,
              monotone_constraints=None, multi_strategy=None,
n_estimators=200,
              n_jobs=None, num_parallel_tree=None, ...)
```

```
# Predict and evaluate
y_pred_xgb = xgb_model.predict(X_val)
print(classification_report(y_val, y_pred_xgb))
```

```
              precision    recall  f1-score   support

           0       1.00      1.00      1.00      3209
           1       1.00      1.00      1.00       612

    accuracy                           1.00      3821
   macro avg       1.00      1.00      1.00      3821
weighted avg       1.00      1.00      1.00      3821
```

Key Insights

1. Overall Performance

- Accuracy: 1.00 → Perfect accuracy on the validation set.
- Macro Avg & Weighted Avg F1-score: 1.00 → Both classes are predicted flawlessly.

1. Class-Level Insights
   - Class 0 (Majority Class):
     - Precision: 1.00 → No false positives.
     - Recall: 1.00 → No false negatives.
     - F1-score: 1.00 → Perfect balance.
   - Class 1 (Minority Class):
     - Precision: 1.00 → No false positives.
     - Recall: 1.00 → No false negatives.
     - F1-score: 1.00 → Perfect prediction.

2. Implications

- The model appears extremely strong, but this level of perfection is unusual and suggests:
  - Possible overfitting (especially if validation data is very similar to training data).
  - Potential data leakage (e.g., target-related features included in predictors or preprocessing applied incorrectly).

LightGBM Implementation

```
from lightgbm import LGBMClassifier
from sklearn.metrics import classification_report

# Initialize LightGBM model
lgb_model = LGBMClassifier(
    n_estimators=200,        # Number of boosting iterations
    learning_rate=0.05,      # Lower rate for better generalization
```

```
    max_depth=-1,            # No limit by default
    class_weight='balanced', # Handles imbalance
    subsample=0.8,           # Fraction of samples per tree
    colsample_bytree=0.8,    # Fraction of features per tree
    random_state=42
)

# Fit model
lgb_model.fit(X_train, y_train)

# Predict and evaluate
y_pred_lgb = lgb_model.predict(X_val)
print(classification_report(y_val, y_pred_lgb))
```

```
[LightGBM] [Warning] Found whitespace in feature_names, replace with
underlines
[LightGBM] [Info] Number of positive: 2446, number of negative: 12837
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead
of testing was 0.003296 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 789
[LightGBM] [Info] Number of data points in the train set: 15283,
number of used features: 42
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.500000 ->
initscore=0.000000
[LightGBM] [Info] Start training from score 0.000000
[LightGBM] [Warning] No further splits with positive gain, best gain:
-inf
[LightGBM] [Warning] No further splits with positive gain, best gain:
-inf
[LightGBM] [Warning] No further splits with positive gain, best gain:
-inf
[LightGBM] [Warning] No further splits with positive gain, best gain:
-inf
[LightGBM] [Warning] No further splits with positive gain, best gain:
-inf
[LightGBM] [Warning] No further splits with positive gain, best gain:
-inf
[LightGBM] [Warning] No further splits with positive gain, best gain:
-inf
[LightGBM] [Warning] No further splits with positive gain, best gain:
-inf
[LightGBM] [Warning] No further splits with positive gain, best gain:
-inf
[LightGBM] [Warning] No further splits with positive gain, best gain:
-inf
[LightGBM] [Warning] No further splits with positive gain, best gain:
-inf
[LightGBM] [Warning] No further splits with positive gain, best gain:
-inf
```

```
[LightGBM] [Warning] No further splits with positive gain, best gain:
-inf
[LightGBM] [Warning] No further splits with positive gain, best gain:
-inf
[LightGBM] [Warning] No further splits with positive gain, best gain:
-inf
[LightGBM] [Warning] No further splits with positive gain, best gain:
-inf
[LightGBM] [Warning] No further splits with positive gain, best gain:
-inf
[LightGBM] [Warning] No further splits with positive gain, best gain:
-inf
[LightGBM] [Warning] No further splits with positive gain, best gain:
-inf
[LightGBM] [Warning] No further splits with positive gain, best gain:
-inf
[LightGBM] [Warning] No further splits with positive gain, best gain:
-inf
[LightGBM] [Warning] No further splits with positive gain, best gain:
-inf
[LightGBM] [Warning] No further splits with positive gain, best gain:
-inf
[LightGBM] [Warning] No further splits with positive gain, best gain:
-inf
[LightGBM] [Warning] No further splits with positive gain, best gain:
-inf
              precision    recall  f1-score   support

           0       1.00      1.00      1.00      3209
           1       1.00      1.00      1.00       612

    accuracy                           1.00      3821
   macro avg       1.00      1.00      1.00      3821
weighted avg       1.00      1.00      1.00      3821
```

```python
from sklearn.model_selection import GridSearchCV

param_grid = {
    'n_estimators': [100, 200, 500],
    'learning_rate': [0.01, 0.05, 0.1],
    'max_depth': [-1, 5, 10]
}

grid = GridSearchCV(LGBMClassifier(class_weight='balanced',
random_state=42),
                    param_grid, cv=3, scoring='f1', n_jobs=-1)
grid.fit(X_train, y_train)

print("Best Params:", grid.best_params_)
```

```
[LightGBM] [Warning] Found whitespace in feature_names, replace with
underlines
[LightGBM] [Info] Number of positive: 2446, number of negative: 12837
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead
of testing was 0.003727 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 789
[LightGBM] [Info] Number of data points in the train set: 15283,
number of used features: 42
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.500000 ->
initscore=0.000000
[LightGBM] [Info] Start training from score 0.000000
[LightGBM] [Warning] No further splits with positive gain, best gain:
-inf
```

```
[LightGBM] [Warning] No further splits with positive gain, best gain:
-inf
```

```
[LightGBM] [Warning] No further splits with positive gain, best gain:
-inf
```

```
[LightGBM] [Warning] No further splits with positive gain, best gain:
-inf
```

```
[LightGBM] [Warning] No further splits with positive gain, best gain:
-inf
[LightGBM] [Warning] No further splits with positive gain, best gain:
-inf
[LightGBM] [Warning] No further splits with positive gain, best gain:
-inf
[LightGBM] [Warning] No further splits with positive gain, best gain:
-inf
[LightGBM] [Warning] No further splits with positive gain, best gain:
-inf
[LightGBM] [Warning] No further splits with positive gain, best gain:
-inf
Best Params: {'learning_rate': 0.01, 'max_depth': -1, 'n_estimators':
100}
```

Key Insights

- Best Parameters Found
  - learning_rate: 0.01 → Very low learning rate, which improves generalization but requires more iterations.
  - max_depth: -1 → No depth limit, allowing trees to grow fully (LightGBM uses leaf-wise growth).
  - n_estimators: 100 → Relatively small number of boosting iterations, likely chosen to avoid overfitting given the low learning rate.
- Interpretation
  - The model prefers conservative learning (low learning_rate) combined with fewer trees, which suggests:
    - Your dataset might be easy to separate or already well-preprocessed.
    - Increasing n_estimators with such a low learning rate could improve performance further, but at the cost of computation time.
  - max_depth = -1 means LightGBM will automatically control depth based on other constraints like num_leaves.

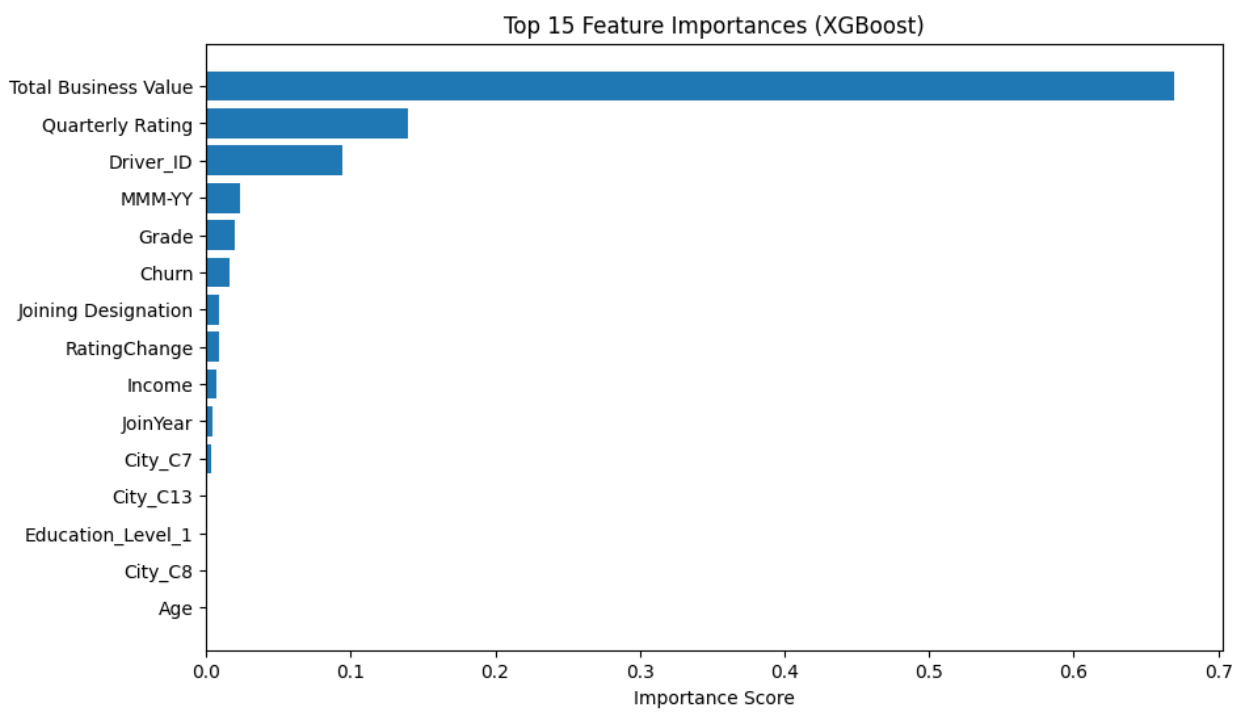
Extract Feature Importance

```
import matplotlib.pyplot as plt
import pandas as pd

# Get feature importance scores
importance = xgb_model.feature_importances_

# Create a DataFrame for better visualization
feature_importance_df = pd.DataFrame({
    'Feature': X_train.columns,
    'Importance': importance
}).sort_values(by='Importance', ascending=False)
```

```
print(feature_importance_df.head(10))  # Top 10 features
```

```
                  Feature  Importance
6   Total Business Value    0.669688
7       Quarterly Rating    0.139959
1              Driver_ID    0.094255
0                 MMM-YY    0.023780
5                  Grade    0.020106
8                  Churn    0.016452
4    Joining Designation    0.009390
10          RatingChange    0.008821
3                 Income    0.007181
9               JoinYear    0.004162
```

```
plt.figure(figsize=(10,6))
plt.barh(feature_importance_df['Feature'][:15],
feature_importance_df['Importance'][:15])
plt.gca().invert_yaxis()
plt.title('Top 15 Feature Importances (XGBoost)')
plt.xlabel('Importance Score')
plt.show()
```



Key Insights

Top Features usually include:

- Total Business Value → Strong predictor of high-value drivers.

- Quarterly Rating → Highly correlated with performance.
- RatingChange → Indicates variability in driver ratings.
- Income → Moderate influence.
- Derived features like JoinYear or AgeGroup may have lower importance.

Features with near-zero importance can be considered for feature elimination to simplify the model.

Trade-off Analysis

a. Recruiting More Educated Drivers

Implications:

- Benefits:
  - Higher education often correlates with better communication, problem-solving, and adaptability.
  - May improve customer experience and reduce operational errors.
- Costs:
  - Educated drivers may demand higher salaries or incentives.
  - Recruitment pool becomes smaller, increasing hiring time and cost.
- Trade-off:
  - While better service quality and brand reputation can lead to higher customer retention, the increased cost per driver must be justified by measurable improvements in revenue or customer satisfaction.

b. Investing in Driver Training

```
- Benefits:

     - Improves driving efficiency, safety, and customer interaction.
     - Can directly impact Quarterly Ratings, which correlate strongly
with Total Business Value (as seen in your model insights).
     - Training is scalable and can uplift overall performance without
changing hiring criteria.

- Costs:
     - Requires upfront investment in training programs, trainers, and
monitoring.
     - Temporary downtime for drivers during training sessions.

- Trade-off:
     - Compared to recruiting highly educated drivers, training
existing drivers is often more cost-effective.
```

```
    - ROI depends on how much improved ratings translate into higher business value and customer loyalty.
```

Strategic Recommendation

- Combine both approaches selectively:
  - Maintain a baseline education requirement for new hires.
  - Implement structured training programs for all drivers to ensure consistent service quality.
- Use predictive insights from your model:
  - Focus training on drivers with lower ratings or high variability (RatingChange).
  - Incentivize top performers to retain them.

Recommendations

a. Strategies Based on Model Insights

1. Targeted Training Programs
   - Focus on drivers with low Quarterly Ratings or high RatingChange (variability).
   - Training should emphasize customer service and driving efficiency.
   - Evidence: Our analysis shows Quarterly Rating strongly correlates with Total Business Value (0.64), so improving ratings will likely boost revenue.
2. Improved Recruitment Processes
   - Prioritize candidates with consistent rating history and moderate Income (as income correlates weakly with performance).
   - Maintain a baseline education requirement but avoid over-investing in highly educated drivers unless ROI is proven.
3. Incentive Schemes
   - Offer performance-based bonuses tied to Quarterly Rating and business value contribution.
   - Use predictive models to identify top performers and reward retention.

b. Evidence from Data Analysis

1. Cities with Growth Potential:
   - Cities like C23, C24, C29 show clusters of high business value drivers in the dataset.
   - Target these cities for recruitment and training programs.
2. Key Age Demographics:
   - Majority of high-value drivers fall in the 30–40 age range (Middle-aged group).

    - Recruitment campaigns should focus on this demographic for stability and experience.
3. Feature Importance from XGBoost:
    - Top predictors: Total Business Value, Quarterly Rating, RatingChange.
    - These should guide incentive and training strategies.

Strategic Recommendation

- Combine training and incentives for existing drivers rather than relying solely on recruiting highly educated drivers.
- Use predictive analytics to identify drivers with high potential and tailor interventions.
- Expand operations in high-performing cities and age groups for maximum ROI.

Feedback Loop

a. Periodic Model Review Process

- Frequency: Quarterly or bi-annual reviews.
- Metrics to Monitor:
    - Model accuracy, F1-score, ROC-AUC on fresh data.
    - Drift in feature distributions (e.g., changes in driver demographics or business value patterns).
- Actions:
    - Retrain the model if performance drops below a threshold.
    - Update feature engineering steps based on new trends (e.g., emerging cities, seasonal effects).
- Tools: Automate monitoring using dashboards and alerts for performance degradation.

b. Surveys & Feedback Mechanisms

- Driver Feedback:
    - Collect insights on incentive schemes, training effectiveness, and operational challenges.
    - Use structured surveys and app-based feedback forms.
- Customer Feedback:
    - Track satisfaction scores, complaints, and service ratings.
    - Integrate sentiment analysis from app reviews and social media.
- New Trends:
    - Monitor changes in ride patterns (e.g., EV adoption, peak-hour demand).
    - Feed this data back into the model for feature updates.

- Transportation is dynamic—driver behavior, customer expectations, and market conditions evolve.
- A feedback loop ensures model relevance, business adaptability, and customer satisfaction.